【1】自动花洒系统 / Sprinkler system

系统目的：自动感应并控制或扑灭初起火灾，抑制热量释放、保护建筑结构，并为人员疏散争取时间。

Purpose: Automatically detects, controls or extinguishes early fire, suppresses heat generation, protects the building structure and provides evacuation time.

主要部件及功能 / Major components and functions

主要部件 / Major components	功能 / Functions
sprinkler water tank 花洒水箱	储存专用消防用水，供水泵向喷头供水。 Stores the dedicated water supply delivered through the pumps to the sprinkler heads.
Duty, standby and jockey pumps 主泵、备用泵及稳压泵	主泵提供灭火流量及压力；备用泵接替故障主泵；稳压泵维持待命压力，应急发电机提供后备供电。 Duty supplies firefighting flow and pressure; standby replaces a failed duty pump; jockey maintains standby pressure. An emergency generator provides secondary power.
Installation control valve set (alarm valve and main stop valve) 系统控制阀组（报警阀及主截止阀）	主截止阀控制、隔离水源；喷水水流使报警阀开启，并向报警装置供水。 The main stop valve controls and isolates the supply; discharge flow opens the alarm valve and supplies the alarm arrangement.
Riser 立管	沿竖向将水输送至花洒配水管网。 Carries water vertically to the sprinkler distribution network.
Main distribution pipe 主配水管	将立管来水输送至各配水管。 Feeds water from the riser to the distribution pipes.
Distribution pipe 配水管	向各花洒支管分配水。 Distributes water to the range pipes.
Range pipes 花洒支管	安装喷头并向相连喷头供水。 Carry and supply the connected sprinkler heads.
Sprinkler head 花洒喷头	快速反应玻璃球受热至动作温度时破裂，喷头洒水以控制或扑灭初起火灾，并抑制热量释放。 The quick-response glass bulb breaks at its operating temperature, opening the head to control or extinguish early fire and suppress heat release.
WATER MOTOR ALARM GONG 水力警铃	利用报警管路的水流发声，警示花洒系统已动作。 Uses alarm-line water flow to sound a local warning that the sprinkler system has operated.
sprinkler inlets 花洒系统入水口	供消防人员接入外部加压水源，补充系统供水。 Allow firefighters to supplement the system from an external pressurised supply.

火灾时的运行逻辑 / Operation during fire

步骤 1 / Step 1：受热开启 / Heat operation

湿式系统内保持充水。火灾热量使附近喷头的玻璃球破裂，受热喷头开启喷水；未达到动作温度的喷头保持关闭。

The wet-pipe system is water-filled. Fire heat breaks a nearby head’s glass bulb, releasing water; heads below their operating temperature remain closed.

步骤 2 / Step 2：供水及报警 / Water supply and alarm

喷水产生水流及压降，主泵投入维持供水，故障时由备用泵接替；水流经报警阀后驱动水力警铃，并向消防控制及指示板反映系统动作。

Discharge and pressure loss bring the duty pump into operation, with standby backup. Flow through the alarm valve operates the gong and reports system operation to the fire control and indicator panel.

步骤 3 / Step 3：持续控制火势 / Sustained fire control

水持续输送至已开启的喷头，冷却、控制或扑灭火灾，为人员疏散争取时间；正常电源失效时由后备电源支持泵组。

Water continues through operated heads to cool, control or extinguish the fire and provide evacuation time; secondary power supports the pumps on normal-power failure.

【2】消防栓／消防喉轆系统
Fire hydrant / hose reel system

系统目的：提供即时的楼内消防供水：喉轆供使用者扑救小火，消防栓供消防人员灭火。

Purpose: Provides immediate internal water supplies: hose reels let occupants tackle small fires, while hydrants support offensive firefighting by FSD personnel.

主要部件及功能 / Major components and functions

主要部件 / Major components	功能 / Functions
fire service (FS) water tank 消防水箱	储存灭火用水，为消防栓及喉轆提供即时水源。 Stores water for immediate hydrant and hose reel firefighting supply.
Fixed fire pumps for hydrant & hose reel system 消防栓／喉轆系统固定消防泵	由手动火警按钮启动，主泵提供流量及压力，备用泵提供后备；配自动后备电源。 Start from the manual call point; duty supplies flow and pressure, with standby backup and an automatic secondary power source.
jockey pump 稳压泵	水箱低于最高消防栓时配置；压力降至所需系统压力的 95%时启动、恢复至100%时停止，消防泵运行时停止。 For tanks below the highest hydrant: starts at 95% of required system pressure; stops at 100% or when fire pumps operate.
Hydrant riser with venting and draining 附排气及排水的消防立管	向各层输水；排气设施排除积气，排水设施放空管内水。 Carries water to the floor outlets; venting releases trapped air and draining empties the pipework.
Hydrant outlet 消防栓出水口	供消防人员接驳消防水带进行灭火。 Connects firefighters' hoses for firefighting.
Pressure reducing hydrant outlet 减压消防栓出水口	静压或泵压过高时降低出水压力，便于安全操作水带。 Reduces excessive static or pump pressure at the outlet for safe hose operation.
Hose reel **Hose reel nozzle** 消防喉轆；喉嘴	喉轆供使用者扑救小火；喉嘴控制并引导水流。 The reel supplies hose for occupants tackling small fires; the nozzle controls and directs the jet.
Fire services inlet 消防入水口	供消防车补充、加压供水；止回阀防止倒流。 Accepts supplementary pressurised supply from a fire appliance; non-return valves prevent backflow.
Intermediate booster pump 中途加压泵	较高系统需要时增压；消防入水口设启停控制、指示及报警蜂鸣器。 Boosts pressure where required; start/stop controls, indicators and an alarm buzzer are provided at the FS inlet.
manual call points 手动火警按钮	喉轆处的按钮启动消防泵及声／光警报。 At hose reel points, initiate the fire pump and audio/visual warnings.
central FS panel 中央消防控制屏	集中显示消防泵运行、故障及电源接通状态。 Centrally indicates pump running, fault and power-on status.

火灾时的运行逻辑 / Operation during fire

步骤 1 / Step 1：报警及启泵 / Alarm and pump initiation

系统平时保持充水。火灾时操作喉轆处的手动火警按钮，发出警报并启动主消防泵；消防泵运行时稳压泵停止。

The system remains primed. Operating the hose reel call point initiates the alarm and duty fire pump; the jockey pump stops when the fire pump operates.

步骤 2 / Step 2：供水灭火 / Water delivery

水经水泵及立管输送至各层。使用者利用喉轆扑救小火，消防人员接驳消防水带灭火；消防车可经消防入水口供水，需要时由中途加压泵增压。

Pumps and risers supply floor outlets. Occupants use hose reels for small fires; firefighters use hydrant hoses and may supply the inlet from a fire appliance, with intermediate boosting where required.

步骤 3 / Step 3：后备及停泵 / Backup and stopping

主泵因电气或机械故障在 15 秒内未能运行时，备用泵自动接替。固定消防泵只能在就地控制屏人工复位停泵，中央消防控制屏持续显示泵况。

If electrical or mechanical failure prevents duty-pump operation within 15 seconds, standby takes over. Fixed fire pumps stop only by manual reset at the local control panel; central monitoring continues.

【3】自动火警探测及警报系统
Automatic fire detection and alarm system

系统目的：迅速自动探测烟、热或火焰等火灾迹象，警告人员疏散，并启动相关紧急联锁控制。

Purpose: Detects fire signs such as smoke, heat or flames early, warns occupants to evacuate and activates relevant emergency controls.

主要部件及功能 / Major components and functions

主要部件 / Major components	功能 / Functions
Heat Detector 热感探测器	按类型响应固定温度或温升速率而报警，适用于容易产生不必要警报的区域。 Responds to preset temperature or temperature rise, depending on type; suitable for areas vulnerable to unwanted alarms.
Ionization Smoke Detector 离子烟感探测器	以检测及参考两个电离室探测烟雾；烟雾改变电离电流，信号积分减少误报，指示灯显示报警。 Uses detection and reference ionization chambers; smoke changes the current. Signal integration reduces unwanted alarms, and an indicator lamp shows operation.
Optical Smoke Detector 光电烟感探测器	以红外发光二极管及光电二极管探测可见烟粒散射光，报警并亮起指示灯。 Uses an infra-red LED and photo diode to detect light scattered by visible smoke particles, signal fire and illuminate its alarm indicator.
manual call points 手动火警按钮	供人员发现火灾时手动发出火警信号。 Allow a person discovering fire to initiate an alarm manually.
Fire control and indicator panel 消防控制及指示板	设于消防控制中心，集中监控警报、花洒、通风空调及消防栓／喉轆系统，设报警指示及复位按钮。 In the fire control centre, centrally monitors alarms, sprinklers, ventilation/AC and hydrant/hose reel systems, with alarm indicators and reset buttons.
audio / visual alarm devices 声／光警报装置	以声音及闪灯警告人员；配设疏散广播时，指示人员前往出口及不要使用升降机。 Warn occupants by sound and flashing lights; advisory audio, where provided, directs occupants towards exits and away from lifts.
Alarm direct link with FSD 消防处火警直线	将火警信号直接传送至消防处通信中心。 Transmits fire alarm signals directly to the Fire Services communication centre.
battery charger set 电池充电器组	保持备用电池充电，由电池在主电源失效时维持控制及报警系统供电。 Keeps standby batteries charged so they sustain the control and alarm system on mains failure.

火灾时的运行逻辑 / Operation during fire

步骤 1 / Step 1：探测及接收 / Detection and reception

探测器感应烟或热并自动发出信号；人员亦可操作手动火警按钮。消防控制及指示板接收信号并显示有关报警位置或区域。

Detectors sense smoke or heat and signal automatically; manual call points provide another input. The fire control and indicator panel receives the signal and indicates its location or zone.

步骤 2 / Step 2：警告及通报 / Warning and transmission

系统启动有关声／光警报，警告人员疏散，并经消防处火警直线传送火警；配设疏散广播时，引导人员前往出口。

The system activates relevant audio/visual warnings for evacuation and transmits the alarm through the direct link to FSD; advisory messages guide occupants to exits where provided.

步骤 3 / Step 3：联锁及持续监控 / Interlocks and monitoring

预设联锁控制有关通风／空调、烟闸及已连接的灭火、门禁设备。控制板持续监控，主电源失效时由备用电池维持运行。

Configured interlocks control relevant ventilation/AC, smoke dampers and connected suppression/access-control equipment. Monitoring continues, with standby batteries sustaining operation on mains failure.

【4】消防员升降机 / Fireman's lift

系统目的：在火灾时为消防员及其装备提供垂直运输，支持消防和救援工作。

Purpose: Provides vertical transport for firefighters and their equipment during fire to support firefighting and rescue.

主要部件及功能 / Major components and functions

主要部件 / Major components	功能 / Functions
Car 轿厢	运载消防员及消防、救援装备前往所需楼层。 Carries firefighters and firefighting/rescue equipment to the required floors.
Controller 控制器	协调消防运行、停车、平层及门操作，使消防员控制升降机而不受普通召梯干扰。 Coordinates firefighter-controlled travel, stopping, levelling and doors without interference from ordinary calls.
Drive motor 驱动电动机	提供动力，经曳引轮及悬吊绳带动轿厢升降。 Supplies power through the drive sheave and suspension ropes to move the car vertically.
Brake 制动器	停止并保持轿厢静止；弹簧施闸、电力释放，断电时施闸。 Stops and holds the car; spring-applied and electrically released, so loss of power applies it.
Door operator 门机	在轿厢停靠楼层时驱动轿门，并联动层门开闭。 Operates car doors and couples to landing doors when the car is at a landing.
Car doors 轿门	封闭轿厢入口，运行时保持关闭并防止意外开启。 Guard the car entrance and remain closed against accidental opening during travel.
Landing doors 层门	遮护各层井道入口，联锁保证门关闭锁紧后才允许运行。 Guard landing openings; interlocks allow travel only with the doors closed and locked.
Emergency Generator 应急发电机	专供消防设施，在正常电源失效时支持消防员升降机等消防设施运行。 Serves fire installations only and supports the fireman's lift and other FSI on normal-power failure.
Cable joining Fireman's lift to essential switchboard 连接消防员升降机与必要配电板的电缆	向升降机输电，以规定耐火性能支持火灾期间的供电连续性。 Supplies the lift from the essential switchboard, with specified fire resistance supporting circuit integrity during fire.

火灾时的运行逻辑 / Operation during fire

步骤 1 / Step 1：消防接管 / Firefighter takeover

消防运行控制启动后，控制器取消普通召梯响应，将轿厢召回指定消防员入口层，供消防员接管及进行消防、救援。

When fire operation is initiated, the controller cancels ordinary call responses and recalls the car to the designated fire service access level for firefighters to take control.

步骤 2 / Step 2：消防员控制运行 / Firefighter-controlled travel

消防员选择所需楼层，控制器协调升降、制动及平层；普通层站召梯不干扰消防运行。门联锁保持有效，到站停车后由消防员控制开门。

Firefighters select floors while the controller coordinates travel, braking and levelling. Ordinary landing calls do not interfere; door interlocks remain effective, with firefighters controlling door opening after stopping.

步骤 3 / Step 3：后备供电 / Emergency supply

正常电源失效时，应急发电机提供后备电力；连接必要配电板的耐火电缆支持火灾期间所需的供电连续性。

On normal-power failure, the Emergency Generator supplies backup power; the fire-resistant connection to the essential switchboard supports supply continuity during fire.